MEHDI BELHAJ

Power Electronics & Control Engineer

High-voltage converters · Motor control · Board design · Aerospace

Toulouse 31300, France · **+33 7 66 55 74 63** · **belhaj.elmehdi.98@gmail.com**
linkedin.com/in/mehdi-belhaj-eea · mehdi-belhaj-eea.github.io · github.com/mehdi-belhaj-eea

`270 / 540 VDC · SiC` `DO-160G · DAL A` `PSIM · LTspice` `MATLAB / Simulink`



AVAILABILITY	EXPERIENCE	CONTRACT	MOBILITY	WORK PERMIT	DRIVING LICENCE	ENGLISH
Immediate	3 years	Permanent · Fixed-term · Umbrella	All of France	Valid to Nov. 2028	B	C1 · fluent

PROFILE

Power electronics engineer (MSc in EEA) with 3 years of experience in aerospace (Safran, Airbus) and high-voltage R&D (SuperGrid Institute). I design high-voltage converters — **Flyback supplies, a 10 kW SiC inverter, EMC filters** — and model their control and that of electric motors (**Simulink, PSIM**), from specification to DO-160G justification and test plans, up to DAL A.

SKILLS

POWER ELECTRONICS	**Flyback**, Buck, SEPIC, LDO · **Three-phase SiC inverters** · Isolated gate drivers (DESAT, soft turn-off) · Snubbers and clamps · Precharge and inrush limiting · **Transformers specified for suppliers**
CONTROL ENGINEERING	**MATLAB / Simulink / Simscape · PI / PID controllers** (pole placement, margins) · Nested current, speed and position loops · Stability (Bode, Nyquist) · BLDC and induction machines
EMC & QUALIFICATION	**DO-160G**: 270 VDC power input (Section 16 cat. D), conducted emissions (Section 21 cat. H), lightning (Section 22 cat. A3), ESD (Section 25 cat. A), environment (Sections 4 and 5) · Damped common- and differential-mode filters · Insulation at altitude (IEC 60664-1) · Dielectric testing
BOARD DESIGN	**OrCAD Capture CIS**, KiCad (3D pre-placement) · **IPC-2221** layout rules · Mechanical integration (Creo View, eDrawings) · Component selection, part numbering and derating · DFX
SIMULATION & TOOLS	**PSIM, LTspice**, Saber, Scilab / Xcos, Mathcad · Python, C++ · Oscilloscope · Windchill PLM

PROFESSIONAL EXPERIENCE

Power Electronics & Hardware Engineer — Safran Ventilation Systems

`Nov. 2023 – May 2025 · 19 months`

Via Centum T&S · Blagnac, France · EMC DO-160G category H

270 VDC mock-up of an aircraft fan originally designed for 115 VAC: 1,890 W input board (−40 to +100 °C) supplying the inverter and the motor control boards.

- **Design**: 9 justified functions — reverse-current protection, precharge (DO-160G inrush envelope), phase-loss detection in under 4 ms, 200 ms power hold-up, lightning (cat. A3), 15 kV ESD, EMC filter.
- **Isolated Flyback supply**: multi-output (132 kHz) from the 270 VDC bus, two isolation groups; 5, 16 and 18 V rails via LDOs and a 1.5 MHz Buck; transformer specification (STB) written for the supplier.
- **EMC filter**: damped differential- and common-mode stages, sized by PSIM simulation of the 20 kHz PWM inverter and the motor (FFT against the category H mask), checked in LTspice (425 V surge, category D).
- **6-layer PCB**: layout specification derived from the mechanical interface drawing and 3D model — IPC-2221 clearances corrected for altitude, DC-bus traces rated 9 A RMS, DFX.
- **Justification and verification**: DJD, calculation notes, HW and EMC test plans, automated test bench instruction (ITE), peer review; weekly meeting in English with Safran USA.

Tools: PSIM · LTspice · Mathcad · Excel (calculation notes) · OrCAD Capture CIS · Windchill PLM · Creo View

Results: 270 VDC feasibility demonstrated with minimal impact on software and mechanics.

Power Electronics & Control Engineer — Airbus / Crouzet

`Aug. 2022 – Nov. 2023 · 16 months`

Via Serma Ingénierie · Toulouse · R&D, Zero Emission programme · DAL A · ABD0100

Control electronics of an electromechanical spoiler actuator deployed in under 130 ms against gusts: 3 stacked boards (CPU, HVDC, inverter), target electronics volume 65 × 65 × 170 mm.

- **Inverter**: three-phase 10 kW SiC inverter for a BLDC motor (6 × 1200 V MOSFETs, phase-current sensing); UCC21750 drivers sized (gate current, DESAT, soft turn-off); loss calculation versus junction temperature redone.
- **Control**: Simulink model of the BLDC motor, inverter and control (Hall-sensor commutation, position / current loops), mechatronic modelling of the actuator, PI tuning.

- **±270 V HVDC board**: EMC filtering, ESD protection, current / voltage sensing, quasi-resonant Flyback 540 V to 24 V, Buck 24 V to 12 V; 2 transformers specified for the supplier.
- **Mechanical integration**: 3D pre-placement of the 3 boards in KiCad with the architect and the mechanical engineers (outline, fixings, aluminium base plate).
- **Specification and safety**: contribution to the requirements specification, sub-assembly specifications, justification files, HW and EMC test plans; requirements with the Safety team.

Tools: MATLAB / Simulink / Simscape · Scilab / Xcos · PSIM · LTspice · OrCAD · KiCad (3D)

Results: 130 ms requirement validated in simulation; hardware architecture approved in Airbus and Crouzet design reviews.

Final-year internship, R&D power electronics — SuperGrid Institute `Feb. – Jul. 2022 · 6 months`
Villeurbanne, France · research institute for HVDC grids

Isolated power supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs.

- **Design**: cascade of 12 ferrite transformers (TDK EPCOS T38), trade-off between losses, insulation and size; model validated with oscilloscope measurements.
- **EMC and testing**: common-mode currents under 16 kV/µs edges and countermeasures; dielectric testing.

Tools: LTspice · oscilloscope · high-voltage dielectric test bench

Results: 20 kV insulation validated, industrialisation specifications delivered.

PERSONAL & UNIVERSITY PROJECTS

- **Personal projects on GitHub (2026)**: **hardware-design-kit**, power-side design calculators for a board (precharge, DO-160G lightning, EMI filter, flyback, measurement chains, PCB rules; 76 tests, ngspice cross-checks); **bldc-control-sim**, field-oriented and six-step control of a BLDC motor on a simulated inverter, C99 code tested against its Python reference. **Tools**: Python · C99 · ngspice · PyTest
- **Induction machine control** (MSc EEA, 2020): simulation of the machine and its control, nested current, flux and speed loops, limiters, robustness tests. **Tools**: Saber
- **BLDC motor on FPGA** (ENSEEIHT, 2021): commutation and PWM modelled, PWM and speed measurement coded in VHDL on a DE0-Nano board (Cyclone IV), NIOS II processor, control law discretised and coded in C with interrupts; speed control validated on a real motor. **Tools**: PSIM · MATLAB · Quartus · Qsys

ENTREPRENEURSHIP

Co-founder — MANMAFOX SAS `Since Jul. 2023`
TokyoSeat smart toilet seats · Amazon EU · side activity

- Product compliance with the manufacturer (China) and an ISO/IEC 17025-accredited test laboratory (CNAS): tests to EN 60335 (safety), EMC, radio, RoHS and ecodesign; analysis of test reports (standards, models covered, conditions, limits); CE (LVD, EMC, RED) and UKCA marking, WEEE obligations, technical files and declarations of conformity.

EDUCATION & CERTIFICATIONS

Master's degree in EEA — Power Electronics, Actuators & Control · Université Toulouse III Paul Sabatier · INP-ENSEEIHT `2020 – 2022`

Bachelor's degree in EEA · Université Toulouse III Paul Sabatier `2019`

Two-year technical degree (DUT) in Electrical Engineering and Industrial Computing · ENSAM Rabat, Morocco `2018`

C++ and Python certifications — Cisco Networking Academy / OpenEDG `2025 – 2026`
C++ Essentials 1 and 2, C++ Advanced, Python Essentials 1 and 2 · 330 h

French electrical safety authorisation B0, H0, B2V, B2V Essai, BE Essai — CAMIRA Formation `2022 – 2025`

EYE Berlin — European Youth Event, European Parliament · certificate of participation `Apr. 2024`

LANGUAGES & INTERESTS

French bilingual · **English** C1, professional use · **Spanish, Portuguese** basic

Autonomous · rigorous · methodical · team player · Competitive football · NQT mentoring programme · Travel (18 countries)